

Supply Chain Analytics on Databricks

This project demonstrates a robust, enterprise-grade Databricks Lakehouse implementation. By unifying disparate supply chain data into a governed Unity Catalog environment, it provides the clean data foundation required for real-time operational KPI tracking, self-service BI, and predictive AI optimization.

01 BUSINESS CHALLENGE

EXECUTIVE BRIEF

- Data Fragmentation:** Highly complex, isolated storage silos across disparate supplier, warehouse, and purchase order fulfillment systems.
- Governance Bottlenecks:** Lack of historical supplier performance changes (SCD) and heavy manual reconciliation dependencies.
- Value At Risk:** High operational friction delays strategic execution, directly lowering reactive supply chain agility and customer service standards.

02 SOLUTION OVERVIEW

LAKEHOUSE PLATFORM

- Medallion Ingestion:** Continuous JSON & CSV pipeline setup through streaming Auto Loader frameworks.
- Change Data Capture (CDC):** Incremental updates processed with structured MERGE and SCD Type 2 performance audits.
- Consolidated Models:** Business KPI gold datasets unified for downstream conversational AI/BI analysis spaces.

03 SOLUTION ARCHITECTURE & DATA PIPELINE



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Project Type: Data Engineering | Medallion Architecture | CDC & SCD Type 2 | Unity Catalog

04 VALUE REALIZATION MATRIX

STRATEGIC OUTCOME

OPERATIONAL METRIC AREA	TARGET STATE IMPACT
Supply Chain Visibility	Improved
Data Freshness / Latency	Increased Velocity
Supplier Performance Insights	Enhanced Auditability
Decision Response Delay	Drastically Diminished

05 STRATEGIC VISION

TRANSFORMATION SCOPE

STRATEGIC SHIFT:

Fragmented Systems

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Unified Lakehouse Intelligence

TARGET INDUSTRIES:

Manufacturing

Retail & E-Commerce

Logistics & Transport

Consumer Goods (CPG)

06 FUTURE SCALE & PRODUCT INTEGRATION ROADMAP



Phase 1: Foundation

Medallion Ingestion & CDC
Pipelines



Phase 2: Operational

Real-Time Dashboards &
Alerts



Phase 3: Predictive

Supplier Risk Models &
Forecasting



Phase 4: Autonomous

Agentic Supply Chain
Optimization

INTEGRATION & ORCHESTRATION SYSTEMS

Delta Live Tables

Unity Catalog Governance

AI/BI Genie

Agentic AI Workflows